



University of Colombo, Sri Lanka



UCSC *University of Colombo School of Computing*

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination — Semester II– UCSC AY22 [held in March/April 2026]

SCS1310 — Object Oriented Modelling and Programming

(Two (2) Hours)

Answer ALL questions

Number of Pages = 16

Number of Questions = 3



238

To be completed by the candidate

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Important Instructions to candidates:

- Students should answer in the medium of English language only using the space provided in this question paper.
- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Write your index number **CLEARLY** on each and every page of this question paper.
- The duration of the paper is **Two (2)** hours.
- This paper has **3** questions in **16** pages (including the Cover Page).
- Answer **all** the questions.
- Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- Do not tear off any part of this question paper. Under no circumstances may this paper, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

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1. Only ONE answer is correct in the following 20 MCQs.

Cross or color the **best-suited answer** among the given options.

Note: Assume that all the other required lines of codes are in place to run all given programs.

[2 x 20 = 40 marks]

(a).	(i)	(ii)	(iii)	(iv)	(v)
(b).	(i)	(ii)	(iii)	(iv)	(v)
(c).	(i)	(ii)	(iii)	(iv)	(v)
(d).	(i)	(ii)	(iii)	(iv)	(v)
(e).	(i)	(ii)	(iii)	(iv)	(v)
(f).	(i)	(ii)	(iii)	(iv)	(v)
(g).	(i)	(ii)	(iii)	(iv)	(v)
(h).	(i)	(ii)	(iii)	(iv)	(v)
(i).	(i)	(ii)	(iii)	(iv)	(v)
(j).	(i)	(ii)	(iii)	(iv)	(v)

(k).	(i)	(ii)	(iii)	(iv)	(v)
(l).	(i)	(ii)	(iii)	(iv)	(v)
(m).	(i)	(ii)	(iii)	(iv)	(v)
(n).	(i)	(ii)	(iii)	(iv)	(v)
(o).	(i)	(ii)	(iii)	(iv)	(v)
(p).	(i)	(ii)	(iii)	(iv)	(v)
(q).	(i)	(ii)	(iii)	(iv)	(v)
(r).	(i)	(ii)	(iii)	(iv)	(v)
(s).	(i)	(ii)	(iii)	(iv)	(v)
(t).	(i)	(ii)	(iii)	(iv)	(v)

MCQs from (a) to (h) are based on the following case study.

CASE STUDY: Online Food Ordering System

A restaurant provides an Online Food Ordering System where customers can browse the menu and place orders. Customers must register and log in before placing an order. After logging in, they can select food items, add them to a cart, and confirm the order. Once the order is confirmed, the system sends the order to the restaurant kitchen for preparation. Customers can make payments through an external payment gateway. A restaurant manager can update the menu by adding, editing, or removing food items.

(a). Which of the following correctly identifies the most appropriate set of actors for the system?

- i. Customer, Restaurant Manager, Payment Gateway
- ii. Customer, Order, Kitchen, Restaurant Manager
- iii. Customer, Cart, Menu
- iv. Restaurant Manager, FoodItem, Order
- v. Customer, Restaurant Manager, Payment, Cart

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- (b). Which UML use-case relationship best models the following requirement?
Customers must log in before placing an order.

- i. Login <<include >> Place Order
- ii. Place Order <<include >> Login
- iii. Login <<extend >> Place Order
- iv. Place Order <<extend >> Login
- v. Association between Login and Order

- (c). Consider the following scenario for creating a corresponding sequence diagram.

- I. Customer → OrderSystem: browseMenu()
- II. Customer → OrderSystem: addToCart()
- III. OrderSystem → Order: calculateTotal()
- IV. Customer → PaymentGateway: processPayment()
- V. PaymentGateway → OrderSystem: paymentConfirmed()
- VI. OrderSystem → Customer: confirmOrder()

Which of the following statements highlights an error in this sequence diagram?

- i. browseMenu () should be called by Order, not the Customer
- ii. calculateTotal () should be invoked by PaymentGateway
- iii. Customer should not interact directly with PaymentGateway
- iv. addToCart () should be an asynchronous message
- v. The diagram is correct and contains no errors

- (d). Consider the following scenario.

- I. The OrderSystem calls processPayment () on the PaymentGateway and waits for the result.
- II. The Order object calls its internal method calculateTotal () to compute the total price of items.

Which of the following correctly identifies the type of messages in the sequence diagram?

- i. processPayment () and calculateTotal () are asynchronous
- ii. processPayment () is synchronous; calculateTotal () is a self-message
- iii. processPayment () is a self-message; calculateTotal () is synchronous
- iv. Both processPayment () and calculateTotal () are return messages
- v. processPayment () is asynchronous; calculateTotal () is synchronous

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- (e). Which of the following best represents the set of core domain classes in the Online Food Ordering System?

- | |
|--|
| i. Customer, Order, OrderItem, FoodItem, Payment |
| ii. Button, Form, Screen, LoginPage, Cart |
| iii. Order, Kitchen, Table, Waiter, Chair |
| iv. Customer, Window, MenuBar, Cart, Session |
| v. Browser, Server, HTMLPage, Response |

- (f). Suppose an Order contains multiple OrderItems, and OrderItems cannot exist without the Order.

Which relationship is most appropriate between Order and OrderItem in the class diagram?

- | | | |
|----------------|-------------------|------------------|
| i. Association | ii. Aggregation | iii. Composition |
| iv. Dependency | v. Generalization | |

- (g). In the Online Food Ordering System, consider the following relationships between Order, OrderItem, and FoodItem.

- An Order may contain multiple OrderItems.
- Each OrderItem belongs to exactly one Order.
- Each OrderItem refers to exactly one FoodItem.
- A FoodItem may appear in many OrderItems across different orders.

Which of the following multiplicities correctly represents the relationship between OrderItem and FoodItem?

- | |
|---------------------------------------|
| i. OrderItem (1) — FoodItem (1) |
| ii. OrderItem (0..*) — FoodItem (1) |
| iii. OrderItem (1) — FoodItem (0..*) |
| iv. OrderItem (1..) — FoodItem (1..*) |
| v. OrderItem (0..1) — FoodItem (1..*) |

- (h). Which of the following statement is correct about modeling Order status in UML?

- | |
|--|
| i. Order status should be an attribute in the Class Diagram, and its changes are shown in a State Diagram. |
| ii. Order status should only appear in the Class Diagram. |
| iii. Order status is only represented in a Sequence Diagram. |
| iv. Order status attribute is a method in the Class Diagram. |
| v. Order status should not be modeled in UML diagrams. |

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- (i). Which statement best describes the difference between a Sequence Diagram and a Communication Diagram?

- i. A Communication Diagram cannot show message order, while a Sequence Diagram shows the message order.
- ii. A Sequence Diagram shows the order of messages over time, while a Communication Diagram shows object relationships with numbered messages.
- iii. A Sequence Diagram shows class inheritance, while a Communication Diagram shows object interactions.
- iv. A Sequence Diagram focuses on databases, while a Communication Diagram focuses on system architecture.
- v. A Sequence Diagram cannot show method calls, but a Communication Diagram shows the method calls.

- (j). Which of the following statements correctly classifies the UML diagrams?

- i. Class Diagram and State Diagram show structure; Sequence Diagram shows behavior.
- ii. Use Case Diagram and Sequence Diagram show structure; Class Diagram shows behavior.
- iii. State Diagram and Sequence Diagram show structure; Class Diagram shows behavior.
- iv. Class Diagram and Object Diagram show structure; State and Communication Diagrams show behavior.
- v. All UML diagrams show only structure; behavior is not modeled.

- (k). Consider the relationship between the objects in the following pairs. Which of them have a relationship defined as Inheritance in OOP?

- I. (Computer, Keyboard)
- II. (Phone, Android)
- III. (Database, MySQL)

Which of the above statement(s) is/are TRUE?

- | | | |
|---------------|--------------------|-----------------------|
| i. I only. | ii. I and II only. | iii. II and III only. |
| iv. III only. | v. I, II, and III. | |

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(l). Which of the following statements is TRUE according to the class definition below?

```
class A : protected B, private C { };
```

- i. All public and protected data members in C will become private in A.
- ii. All public, protected, and private data members in B will become protected in A.
- iii. Both private and protected data members in B will become protected in A.
- iv. Both public and protected data members in B will become protected in A.
- v. Only A can access any data members in C.

(m). What is the output of the following program written in C++?

```
class A{
public:
    A() {cout<<"X";}
    A(int x) {cout<<x;}};

class B: public A {
public:
    B() {cout<<"Y";}
    B(int x) {cout<<x;}};

int main(){
    B obj1;
    B obj2(10); }
```

- | | | |
|-----------|------------|------------|
| i. XY10 | ii. XY10X | iii. XYA10 |
| iv. YXX10 | v. YXX1010 | |

(n). Consider the following three statements regarding Abstraction in OOP.

- I. It prevents unauthorized access to the data in objects.
- II. It is achieved by overriding functions.
- III. It can be implemented using pure virtual functions.

Which of the above statement(s) is/are TRUE?

- | | | |
|---------------|--------------------|-----------------------|
| i. I only. | ii. II only. | iii. II and III only. |
| iv. III only. | v. I, II, and III. | |

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(o). Consider the following three statements regarding the Function Overloading in C++.

- I. It means redefining a function in a derived class.
- II. It is achieved through compile-time polymorphism.
- III. It requires the functions to have different return types.

Which of the above statement(s) is/are TRUE?

- | | | |
|----------------------|--------------------|---------------------|
| i. I only. | ii. II only. | iii. I and II only. |
| iv. II and III only. | v. I, II, and III. | |

(p). What is the output of the following program written in C++?

```
class A{
public:
    void f(char x, int y=1) {cout<<x;}
    void f(int x, char y) {cout<<y;}
    void f(int x) {cout<<x;}
};

int main() {
    A obj;
    obj.f(65);
    obj.f('A');
    obj.f(65, 'A');
}
```

- | | | |
|----------|----------------------|----------|
| i. 65A65 | ii. 65AA | iii. 65A |
| iv. A65A | v. Compilation Error | |

(q). Consider the following three statements regarding the function definition given below.

```
virtual int f(int x=1) = 0;
```

- I. This function returns 0 when called with no arguments.
- II. You cannot create objects of the class that contains this function.
- III. You must overload this function before using it.

Which of the above statement(s) is/are TRUE?

- | | | |
|----------------------|--------------------|----------------|
| i. I only. | ii. II only. | iii. III only. |
| iv. II and III only. | v. I, II, and III. | |

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(r). Consider the following three statements regarding Composition in OOP.

- I. It is used to build complex objects from simpler objects.
- II. It allows code reuse by using existing classes as components.
- III. The (Teacher, Student) pair has a relationship defined by composition.

Which of the above statement(s) is/are TRUE?

- | | | |
|---------------------|--------------------|---------------|
| i. I only. | ii. I and II only. | iii. II only. |
| iv. I and III only. | v. I, II, and III. | |

(s). Consider the following three statements regarding templates in C++.

- I. They support generic programming.
- II. They are instantiated at compile time.
- III. They are used to pass data values to functions.

Which of the above statement(s) is/are TRUE?

- | | | |
|---------------|--------------------|---------------|
| i. I only. | ii. I and II only. | iii. II only. |
| iv. III only. | v. I, II, and III. | |

(t). Consider the following three statements regarding exceptions in OOP.

- I. It is an error that occurs during the compilation of a program.
- II. It allows a program to handle errors in a controlled way.
- III. It prevents the execution of a program before it causes serious damage.

Which of the above statement(s) is/are TRUE?

- | | | |
|---------------|--------------------|----------------------|
| i. I only. | ii. II only. | iii. I and III only. |
| iv. III only. | v. I, II, and III. | |

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2. Consider the Product class definition given below.

In this class, the member variables **n**, **p**, and **q** represent the **name of the product**, **price of the product**, and **quantity of the product**, respectively. Some member functions are defined within the class definition.

```
class Product {
    string n;
    double p;
    int q;
    static int totProducts;

public:
    Product();
    Product(string name, double price, int qnt) {
        n=name; p=price; q=qnt; }

    Product(const Product &);

    void updateStock(int);
    void updateStock(int, double);

    Product operator + (const Product &);
    bool operator == (const Product &) const;

    static void showInventory() {
        cout << "Total Products: " << totProducts << endl; }

    void display() const {
        cout << "Name: " << n << ", Price: " << p
            << ", Quantity: " << q << endl; }
};

int Product::totProducts = 0;
```

(a). Complete the declarations and implementations of the following constructors.

- i. A default constructor using an initializer list to initialize the member variables **n**, **p**, and **q** to **"Unknown"**, **0.0**, and **0**, respectively.

<pre>..... Product() : { totProducts++; }</pre>
--

[2.5 marks]

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- ii. Copy constructor that creates a new `Product` object by copying another product.

```
..... Product(const Product &prod) {  
  
.....  
  
.....  
  
totProducts++;  
cout<<"Copy Constructor"<< endl;}
```

[2.5 marks]

- (b). The member functions `updateStock` will be used to update the existing stock levels.

- i. Implement `void updateStock(int q)` to increase the product quantity by `q`.

--

[1 marks]

- ii. Implement `void updateStock(int q, double newP)` to increase the quantity by `q` and update the product price to `newP`.

--

[2 marks]

- iii. Identify the object-oriented concept illustrated by these two functions and briefly justify your answer.

--

[2 marks]

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(c). Briefly explain the purpose of the following in the `Product` class.

i. static variable `totProducts`

--

[2.5 marks]

ii. static function `showInventory()`

--

[2.5 marks]

(d). The `Product` class includes two operator member functions.

i. Implement the overloaded operator `+`, where the function should:

- Create and return a new `Product` object.
- Set the name of the new product by combining the names of the two products with " & " between them.
- Set the price of the new product as the average of the prices of the two products.
- Set the quantity of the new product as the sum of the quantities of the two products.

--

[5 marks]

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- ii. Implement the overloaded operator `==` that returns true if two Product objects have the same price, and false otherwise. Briefly explain the purpose of making this function a constant.

--

[3 marks]

- iii. Name the object-oriented concept illustrated by the above implementations. Briefly explain your answer.

--

[2 marks]

- (e). Given the following `main()` function, write the exact output produced by the program when it is executed.

```
int main() {  
    Product p1;  
    Product p2("Pen", 10.0, 50);  
    Product p3("Notebook", 50.0, 30);  
    Product p4 = p2 + p3;  
    p4.display();  
    p1 = p3;  
    p2.updateStock(20);  
    p3.updateStock(10, 55.0);  
    Product p5(p1);  
    p5.display();  
    cout << "Are p1 and p4 same price? "  
    << (p1 == p4 ? "Yes" : "No") << endl;  
    Product::showInventory(); }  
}
```

--

[5 marks]

--	--	--	--	--	--	--	--

3. (a). Write the output of each of the following C++ programs.

If a program resulting a compilation error, write "Compiler Error" as the answer.
Note that all syntax rules (such as proper use of semicolons and balanced brackets) are correct in these programs.

```
i. class A {
    public: A() {cout<<"A";} };

class B {
    public: B() {cout<<"B";}
           B(int x) {cout<<x;} };

class C: public B, public A {
    public: C() {cout<<"C";}
           C(int y) {cout<<y;} };

int main() {
    C c(5);
}
```

[3 marks]

```
ii. class A { public: A(){cout <<"A";} int x; };
class B : public A {public: B(){cout <<"B";}};
class C : public A {public: C(){cout <<"C";}};
class D : public B, public C {public: D(){cout <<"D";}};

int main() {
    D obj;
    obj.x = 10;
    cout<<obj.x;
}
```

[3 mark]

```
iii. class A {
    public:
        void f(int x) {cout<<"X";}
        void f(char y) {cout<<"Y";}
        char f(char z) {return z;}
};

int main() { A a; a.f('A'); }
```

[3 mark]

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iv. `template <typename T, typename U, int n=1>
U f(T x, U y) { return x * y + n; }`

```
int main() {  
    cout << f<char, int>('A', 1);  
    cout << f<int, char, 1>(1, 'B');  
}
```

[3 mark]

.....

v. `template <typename T, typename U>
class A {
 public:
 T f(U x) { return x + 1; }
 U g(T x) { return x + 2; }
};`

```
int main() {  
    A<char, int> a;  
    cout << a.g(65);  
}
```

[3 mark]

.....

vi. `int main() {
 try {
 cout<<"A";
 throw 65;
 cout<<"B";
 }
 catch (char x) {cout<<"C"; }
 catch (int x) {cout<<"D"; }
 catch (...) {cout<<"E"; }
 cout<<"F";
}`

[3 mark]

.....

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```
vii. int main() {
    cout << "V";
    try {
        throw 'a';
    }
    catch (...) { cout << "W"; }
    catch(int x) { cout << "X"; }
    catch(char y) { cout << "Y"; }
    cout << "Z";
}
```

[3 mark]

.....

```
viii. class A {
    public:
        int fa() {return 5;}
};

class B {
    public:
        int fb(int x) {return x*2;}
};

class C {
    private:
        A a; B b;
    public:
        int fc(int x) {
            return b.fb(a.fa()) + x;
        }
};

int main() {
    C c;
    cout << c.fc(15);
}
```

[3 mark]

.....

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(b). Use the following class definition to answer the questions below.

```
class A {  
public:  
    string f1() {return "A";}   
    void f2() { cout<<f1();}  
};
```

```
class B: public A {  
public:  
    string f1() {return "B";}   
};
```

```
int main() {  
    A a; B b;  
    a.f2();  
    b.f2();  
}
```

i. What is the output of the program?

[2 marks]

.....

ii. Briefly explain why this output is produced.

--

[2 marks]

iii. What is the output of the program if we define
A::f1() as a virtual function?

[2 marks]

.....

_____ ***** _____